# Independent Scientific Assessment of Anthropic's Reconstructed Ultraviolet Sky Map

Claims, Validation, Measurement Provenance, Uncertainty, and Reproducibility

**Prepared for** Grounded DI LLC **Review** 09 October 2026 **Version** 1.0 **Status** Source-level; no numerical replay

> **Disposition.** The scientific reconstruction is supported by substantial methodological and internally validated evidence. Its numerical accuracy is **not independently reproduced** by this review; its accuracy in the most difficult genuinely unobserved Galactic-plane conditions remains **unresolved**. Public documentation is notably forthcoming about several of these limitations, although selected headline formulations compress important qualifications.

## 1. Executive assessment

Ménard's all-sky ultraviolet product is a credible scientific reconstruction, not a free-form generated image. It unifies observational inputs from GALEX and Swift-UVOT with a FIMS/SPEAR far-ultraviolet constraint, predicts missing diffuse emission from other astronomical tracers, inserts an extensive catalogue-based stellar layer, and specifies pixel-level provenance and uncertainty products. The manuscript documents spatial cross-validation, source and seam checks, end-to-end mock gaps, instrument comparisons, uncertainty coverage, and important limitations [M, abstract; §§3–4, 6–7].

The principal inference limitation is **transport of performance from observed sky to the sky GALEX did not observe**. Six-fold spatially blocked holdouts are stronger than random pixel tests, but the covariate distribution remains different from the heavily extincted and source-crowded Galactic plane. The paper reports an FUV two-sample classification AUC of 0.83 between matched held-out and filled pixels, a FUV importance-weight effective sample size of 2.2%, and only 110 reported held-out FUV pixels above E(B−V)=1.6. It also states that 8.2% of filled FUV sky falls beyond comparable GALEX support [M, §4.2, pp. 19–20; Appendix B, p. 36]. The forty hidden-block end-to-end experiment does not include a block with E(B−V)>0.5 [M, §4.3, p. 22].

An accessible technical webpage says that downloads of the FITS and HiPS maps will be added; the manuscript describes a detailed public replay archive that is not linked from that page at the review cut-off [T, §6; M, §7, pp. 31–33]. Consequently, reported statistics and FITS integrity were **read and checked for internal consistency**, not independently recomputed from arrays. The most useful next steps are the promised versioned artifact release followed by a blinded UV comparison in representative, high-extinction, GALEX-free sky.

## 2. What the map measures—and what it predicts

The published product is a continuous map of total surface brightness in the GALEX far-UV (FUV, 1350–1750 Å; effective wavelength approximately 1539 Å) and near-UV (NUV, 1750–2800 Å; approximately 2316 Å) bands. Values are expressed

in continuum units (CU) on a Galactic-coordinate HEALPix NESTED grid at `nside=2048`, about 1.7 arcminutes per pixel; a lower-resolution `nside=512` version and uncertainty layers are specified [M, abstract, §§2–3, 7; T, §5]. A full grid is not evidence that every pixel was photographed.

**The evidentiary layers are different:**

1. **Direct ultraviolet imaging:** the GALEX GR6/7 mission supplied the main FUV and NUV footprint; its exposure mosaics underwent foreground subtraction, tile equalization and artifact correction. The final GALEX-dominated value is calibrated and processed imaging, rather than an unchanged telescope count [M, §§3.2–3.4].
2. **Transferred ultraviolet information:** Swift-UVOT observations were calibrated into approximate GALEX-equivalent bands. Where only GALEX NUV was imaged, an NUV-informed FUV layer transfers measured NUV spatial structure through a learned diffuse FUV/NUV colour and adds catalogue-based stars. This is *image-conditioned FUV prediction*, not direct FUV measurement [M, §§3.6, 3.8, 3.10].
3. **Coarse observed constraints:** FIMS/SPEAR FUV spectro-imaging provides a large-scale level in portions of the GALEX-free plane; the smaller-scale texture there is obtained from the predictor and catalogues. FIMS-constrained sky is not equivalent to GALEX-resolution imaging [M, §§3.9–3.10].
4. **Statistical diffuse prediction:** a 90-feature histogram-gradient-boosting model relates star-subtracted GALEX intensity to dust, Hα, Gaia starlight, position and multi-scale derived features, then extrapolates those relationships into gaps. It is a regression using physical tracers, neither direct photon detection nor general-purpose visual image generation [M, §3.7, Table 3].
5. **Synthetic catalogue sources:** the manuscript lists 118,932,137 Gaia stars in its source layer; ultraviolet fluxes for many are inferred from catalogues and stellar predictors, with specified uncertainty [M, §§3.5, 6, Table 7].
6. **Composite visualization:** the seamless full-sky view depicts a combination of these layers. Rendering continuity does not imply observational continuity or a uniform spatial transfer function [A; M, Figs. 1–3, 8].

Other public data include Planck dust, Hα, H I, TD-1 bright-star photometry, Gaia DR3, catalogued galaxies and related calibration and source catalogues. Each has a different role; not all are independent validators simply because they are separate instruments [M, Table 2, §§2–3].

### 2.1 Provenance distribution

The manuscript's dominant-weight classes are: FUV **61.9% GALEX**, **9.7% NUV-informed**, **0.2% UVOT**, **23.7% FIMS-constrained**, **4.5% pure model**; NUV **73.3% GALEX**, **0.2% UVOT**, **26.5% model** [M, §3.9 and Table 4, pp. 16–18]. These fractions are *reported file statistics*, not quantities independently recounted in this review. The paper separately states that predicted sub-degree morphology covers **28.0% FUV** and **26.5% NUV**. Counting image-conditioned FUV transfer as "measured FUV" would obscure its spectral dependence; calling every FIMS-constrained pixel "purely model-generated" would equally misstate its coarse observational support.

## 3. Consolidated findings

### 3.1 Scientific legitimacy and novelty

The methods describe a substantively credible astronomical mosaic-and-reconstruction workflow, with explicit calibration equations, predictor features, fusion weights, benchmarking, uncertainties, source layers and failure cases [M, §§2–7]. However, credibility of the described method is different from independent verification that the released numeric product matches the description.

The unqualified public phrase "first complete map of the sky in UV light" requires revision. **Jo et al. (2021)** published a far-ultraviolet *all-sky* reconstruction using FIMS observations and neural-network predictions for missing coverage [J, abstract and §§2–4]. Ménard's manuscript itself acknowledges Jo et al. and states a narrower contribution: both GALEX bands over the full sphere at common sampling, with per-pixel provenance, uncertainty, a stellar layer and predicted sub-degree

morphology [M, §1.1, p. 4]. The earlier one-band, roughly degree-scale map and the present finer-sampled, two-band product are not the same achievement. The narrow contribution is **SUPPORTED**, while the broad "first complete UV map" phrase is **MISLEADING without qualification**.

### 3.2 Transportability: observed-sky validation versus missing-sky deployment

The six folds are unions of `nside=8` (about 7.3°) super-pixels. Holding out blocks addresses direct spatial-neighbor leakage more rigorously than random isolated pixels. The authors additionally match held-out GALEX pixels to the filled sky on dust reddening E(B−V) and |Galactic latitude| [M, §4.2 and Appendix A]. These are material strengths.

Nonetheless, the held-out pixels and filled pixels do **not** become exchangeable after matching. The manuscript reports FUV/NUV two-sample AUC 0.83/0.66 on additional templates, with FUV differences driven especially by Hα, starlight and source density. Only about 2.2% of the nominal FUV held-out sample survives as Kish effective sample size under importance weighting, and the largest 1% of observations carry 23% of the weight [M, §4.2, p. 19; Appendix B, p. 36]. For 15–49% of model-filled FUV pixels, classifier scores lie above the held-out 99th percentile [M, Appendix B, p. 36]. Such figures demonstrate a statistically meaningful source-to-target mismatch. They do **not** directly measure how wrong the prediction is in the target region.

Distance presents a second deployment shift. Around 25.5% of filled FUV sky lies more than 3.65° from GALEX and 5% more than 8° away. The manuscript reports a 0.048 dex matched scatter near training pixels, rising to approximately 0.08 dex at 5–8°, with extrapolated estimates of **0.10–0.16 dex** for its most distant 5% [M, §4.2, pp. 19–20]. Those deepest-gap values are explicitly poorly determined and should not be presented as measured independent errors. NUV has smaller typical holes but still significant covariate shift and sparse direct imaging validation of the plane [M, §6].

The full-fusion experiment hides forty 7.3° blocks, re-estimates associated gain fields, and uses out-of-fold predictors. It reports 0.049/0.047 dex FUV/NUV scatter at `nside=512`, a legitimate internal stress test. It does not include any hidden block with E(B−V)>0.5, a limitation the authors state explicitly [M, §4.3, pp. 20–22]. Reproducing moderate-dust blocks does not independently validate the highest-dust Galactic plane.

### 3.3 What the accuracy numbers actually mean

The public article says predictions for deliberately hidden observed regions were within "about 10%," while the project page describes typical filled-region errors of 12–14% or 0.05–0.06 dex [A; T, §4]. The manuscript usually scores **robust dispersion of log intensity ratios** rather than an arithmetic percentage error or a uniform per-pixel tolerance. Its benchmark defines `r=log10(prediction/reference)` and robust scatter as `1.4826 × weighted median(|r−weighted_median(r)|)` [M, §4.2, Table 5; Appendix A].

For orientation, `10^0.05−1 ≈ 12.2%`, `10^0.06−1 ≈ 14.8%`, and `10^0.10−1 ≈ 25.9%` on the upward side; log errors are asymmetric when converted into ordinary percent deviations. None of these computations transforms a robust scatter into a bound satisfied by every pixel. Accordingly, **"about 10%" is a simplified characterization of comparatively favorable withheld regions, not a demonstrated global bound**.

The manuscript's multiple 0.037–0.060 dex values are largely a function of changed targets, weighting, regions and reduction stages. Section 4 and Table 5 contain fuller benchmark definitions; the numerical register in §4 of this report preserves all unresolved differences rather than averaging them.

### 3.4 Cross-instrument checks and their dependence

GALEX-free FUV comparisons are useful precisely because their targets were not GALEX images. Against the FIMS/SPEAR "starry" map the authors report a ratio around 0.92 and 0.19 dex scatter in approximately 45,354 coarse pixels. But FIMS/SPEAR directly sets the large-scale level in FIMS-dominated fusion regions. The manuscript explicitly disclaims full independence at angular scales above approximately 4° there [M, §4.4, pp. 22–23]. Small-scale comparison components remain informative but do not constitute unconditioned validation of the large-scale level.

In GALEX-free NUV regions, the comparison against calibrated UVOT UVM2 reports a ratio of approximately 1.005 and scatter of 0.146 dex; a low-UVOT-weight subset with less construction dependence shows approximately 0.21 dex scatter [M, §4.4]. For the FUV inner plane, a UVOT-derived FUV reference differs from the adopted FIMS-based level by about 0.2 dex, with additional dust-dependent spectral-transfer uncertainty. This is a scientifically meaningful cross-instrument bracket; it is neither proof of uniform 0.2 dex map error nor an independent vindication of fine-scale morphology [M, Fig. 12, §4.4].

### 3.5 Resolution, reconstruction detail and false precision

Grid sampling at 1.7′ is a storage choice. The project page says the effective information resolution of model-dominated regions is roughly **0.5°–1°**, with FIMS supplying degree-scale FUV levels [T, §5]. The manuscript also refers to an effective predicted diffuse beam of about 4′ while quantifying **13%, 27% and 40% attenuation** at successive angular multipoles and acknowledging residual cell-edge artifacts [M, §§4.7, 6]. Those expressions are not equivalent resolution measurements: nominal sampling, nominal beam/processing scale, recovered angular power and independent predictability must be reported separately.

It is therefore unsafe to treat a compact feature seen only in model/FIMS-dominated sky as newly observed UV structure. The paper itself advises that the reconstructed regions cannot independently support discovery of previously unknown filaments, shells or sources [M, §§3.10, 6; T, §3]. The hazard is **over-reading the visualization**, not evidence that the statistical method is an unphysical image generator.

### 3.6 Pixel provenance and measurement preservation

The manuscript specifies a convex, priority-nested fusion in descending order GALEX → NUV-informed FUV → UVOT → FIMS-constrained model → pure model. Its equations (8)–(9) enforce weights summing to one in theory; dominant-class labels are the `argmax` of those weights [M, §§3.9–3.10]. Under full feathering GALEX can have weight exactly one, and the authors report `W_GALEX>0.99` for 87% of imaged FUV pixels. However, **neither weight normalization nor the class-fraction headers were independently examined in a FITS file here**.

“Nothing measured is overwritten,” as presented on the project page, needs operational qualification. The pipeline subtracts foregrounds, equalizes tiles, applies repair multipliers, feathering and post-fusion edits. The manuscript explicitly describes replacing a handful of GALEX/NUV cold pixels and inpainting a specific glint footprint [M, §3.10, pp. 16–17; T, §3]. The defensible statement is **observed imaging generally receives priority after disclosed processing, subject to specified exceptions**. This does not establish widespread inappropriate alteration.

An important provenance caveat is the default “data-only” FUV product: it allows NUV-informed FUV pixels even though their FUV level and color are predictions. The manuscript recommends a stricter `W_GALEX+W_UVOT≥0.5` selection for genuinely FUV-measurement-conditioned applications [M, §3.10, p. 17]. This is a use-specific classification problem, not a simple binary measured/predicted distinction.

### 3.7 Statistical uncertainty, correlated systematics and tails

The specified two-column `nside=512` uncertainty product separates **SIGMA** (an approximate central 68% log-intensity half-width, computed from weighted layer terms) and **SIGMA_SYS** (a coherent level systematic, e.g., 0.10 dex in the FIMS-constrained plane). Pixel-to-pixel correlation means SIGMA_SYS must not be treated as independent random noise for every pixel [M, §4.6; T, §§3–5]. The inherited UV-BKGD absolute foreground zero-point uncertainty, potentially up to about 100 CU, is not included in these columns [M, abstract; §§3.2, 4.6].

Aggregate held-out SIGMA coverage is close to its intended nominal rate: §4.6 reports 0.677/0.678 area-weighted FUV/NUV coverage. This is a genuine reported strength. But the same section reports **0.52 FUV coverage** for E(B−V)=0.8–1.6, while Table 10 lists **0.54**, with a recommendation to inflate FUV SIGMA by approximately 1.6 in this stratum [M, §4.6, p. 23; Table 10, p. 37]. Extreme standardized residual tails are considerably broader than Gaussian expectations; for example,

the reported FUV 99.73rd percentile of |residual|/SIGMA is about 15 [M, §4.6; Table 10]. Near-68% aggregate calibration is therefore **not** proof of appropriately calibrated local intervals in every hard region.

### 3.8 Reproducibility and AI review

The manuscript is unusually specific about its intended release, naming the FITS maps, weight and uncertainty arrays, stellar catalogues, exact benchmark arrays, scorer, prediction code, fusion code, manifests and header constants. It also distinguishes downstream stages claimed to be fully scripted from upstream processes produced interactively and not regenerable from raw HiPS tiles by a single command [M, §7, Table 7]. This is stronger documentation than an image plus an anecdotal claim, but it is not an executable replay in this review.

As of the review cut-off the public project page's Downloads section says that FITS and HiPS links **will be added**; it links a build-up PDF and the manuscript rather than the named replay archive [T, §§6–7]. Searches for the distinctive benchmark and manifest filenames did not establish a public executable download. We therefore classify all FITS integrity checks, benchmark rescoring, source-code behavior checks and end-to-end replication as **not performed**. The raw observational surveys are separately public through their custodial archives; that does not establish that Ménard's processed layers and scripts have been released.

The Anthropic article describes AI agents conducting technical work and multiple self-reviews and identifies a visible field-footprint error that the human author caught after two agent reviews [A, “Trial and error”]. Such iterative automation plausibly accelerates analysis and debugging, but same-workflow agent agreement is not independent data, source-code provenance or human peer review. The technical website explicitly states that the work has not undergone human peer review [T, §4].

## 4. Numerical and definitional reconciliation register

All numerical values below are **author-reported** unless specifically marked **arithmetic check**. Distinct target definitions must not be combined into one “accuracy” number.

| Numerical statement | Source and precise scoring context | Reconciled interpretation / open question |
|---|---|---|
| **FUV 0.037/0.047; NUV 0.047/0.057 dex** | [M, §§3.7, 4.2, pp. 14 and 19]; star-subtracted *diffuse* target; area-weighted / gap-matched robust scatter, six-fold blocked CV. | Each slash pair is **area/matched** within a band, not FUV/NUV. The 0.047/0.057 **gap-matched diffuse FUV/NUV** pair on the project page follows from the second entries. |
| **FUV 0.041/0.052; NUV 0.047/0.058 dex** | [M, §3.7, p. 14]; prediction plus stellar source layer against equalized total `g` ; area-weighted / gap-matched. | Project page's **0.052/0.058 FUV/NUV** is this gap-matched pair. |
| **FUV/NUV 0.057/0.060 dex** | [M, §4.2, p. 19]; prediction + source against total GALEX, gap-matched. | Not identical to §3.7's 0.052/0.058. The target/preprocessing/sample specification is not explicit enough to prove complete equivalence. **UNRESOLVED minor cross-section difference** pending arrays/score definitions; do not silently replace. |
| **FUV/NUV 0.060/0.060 dex** | [M, Table 5, p. 20; Appendix A]; total-intensity benchmark vs pre-equalization destriped GALEX `G′` , gap matched. | Different target convention from the equalized diffuse benchmark; neither a second independent experiment nor a universal error bound. |
| **0.049/0.047 dex** | [M, §4.3, pp. 20–22]; full-fusion 40-block mock-gap test at `nside=512` . | Stronger pipeline-level holdout; excludes high-column blocks above E(B−V)=0.5, limiting transportability. |

| Numerical statement | Source and precise scoring context | Reconciled interpretation / open question |
|---|---|---|
| **"About 10%" versus "12–14%"** | [A, Building the missing map; T, §4; M, §4.2]. | Broad public shorthand for log-robust scatter in held-out regions. Understates conditional errors in distant, high-dust deployment areas; neither statistic is a maximum percentage error. |
| **0.07–0.10, up to 0.10–0.16 dex** | [M, §4.2, pp. 19–20]; estimates for far-away FUV gap populations. | Estimates are conditional and the deepest 5% is expressly poorly constrained; **not** independently observed target-domain residuals. |
| **110 held-out FUV pixels >1.6 versus no GALEX FUV pixel >1.6** | [M, §4.2, p. 19; §6(i), p. 29]. | Literal statements conflict without a grid, threshold, footprint or mask qualification. Could reflect coarse regridding or small boundary samples, **but no explanation was verified**. Request exact coordinates/masks and raw count. |
| **FUV 0.52 versus 0.54 coverage at E(B−V)=0.8–1.6** | [M, §4.6, p. 23; Table 10, p. 37]. | Likely different subsets/weighting, but the document does not sufficiently explain the difference for independent resolution. Both establish materially subnominal coverage. |
| **~68% aggregate coverage; very heavy tails** | [M, §4.6 and Table 10]. | Aggregate 68% does not establish conditional or 95–99.7% Gaussian calibration. SIGMA_SYS must be handled separately. |
| **1.7′ sampling; ≈4′ beam; 0.5°–1° predictive information; degree-scale FIMS** | [T, §5; M, §§3.10, 4.7, 6]. | Pixel size, nominal beam, supported predictive information and large-scale external constraint are distinct objects; avoid a single "resolution" headline. |
| **0.19 dex FIMS; 0.146 dex UVOT; ~0.2 dex plane bracket** | [M, §4.4 and Fig. 12]. | Cross-instrument scatter/scale uncertainty, with unequal independence; **not** direct intrinsic error of the predictor. |
| **61.9/9.7/0.2/23.7/4.5% and 73.3/0.2/26.5%** | [M, §3.9, Table 4]; dominant provenance classes. | Documented target fractions; verification requires inspecting the complete FITS weight/provenance arrays. |

The two textual inconsistencies about extreme-dust counts and conditional uncertainty coverage should be treated as **clarification requests**, not evidence that the fitted map is false. The most meaningful numerical result is the performance *gradient* with distance and covariate mismatch, not whichever single robust-scatter figure is lowest.

## 5. Evidence and claim matrix

Classification key: **VERIFIED** = independently checked with sufficient evidence; **SUPPORTED** = persuasive source documentation without independent replay; **UNVERIFIED** = insufficient accessible validation; **CONTRADICTED** = demonstrable conflict with sufficient evidence; **MISLEADING** = defensible wording omitting material qualification; **UNRESOLVED** = specified further evidence required. One claim can be split when qualified and unqualified versions deserve different decisions.

| Claim | Primary source | Supporting evidence | Independent verification performed | Classification | Qualification |
|---|---|---|---|---|---|
| Credible multi-survey reconstruction methodology | [M, §§2–4; A] | Instrument inputs, calibration, predictor, fusion and validation are specified. | Cross-checked methodological descriptions. | **SUPPORTED** | Actual output arrays and pipeline were not run. |
| First complete UV map, without qualification | [A; J; M, §1.1] | Jo et al. published a complete FUV map in 2021. | Read earlier peer-reviewed article and manuscript's own distinction. | **MISLEADING** | Narrow to first of the specific two-band, provenance-labelled type. |

| Claim | Primary source | Supporting evidence | Independent verification performed | Classification | Qualification |
|---|---|---|---|---|---|
| Two-band complete product of the described design | [M, §1.1] | Both bands, common HEALPix sampling, source layer, per-pixel provenance. | Checked claims against identified predecessor, not exhaustive literature. | **SUPPORTED** | Historical priority beyond known examples not comprehensively searched. |
| 28.0% FUV / 26.5% NUV predicted sub-degree structure | [M, §3.10, Table 4; T, §3] | Dominant FIMS and model fractions and morphological caveats. | Compared reported fractions across source texts. | **SUPPORTED** | No FITS weight-array recount. |
| 118,932,137 Gaia stars in source layer | [M, §3.5, Table 7] | Source-catalogue count and origin flags specified. | Located repeated numerical assertion, not the source catalogue itself. | **SUPPORTED** | UV flux predicted for many stars. |
| Uniform ~10% reconstruction error throughout missing plane | [A; T, §4; M, §4.2] | Held-out scatter and increasing distant-gap errors documented. | Compared conditions and measures stated in primary texts. | **MISLEADING** | A public shorthand, not a universal validation bound. |
| Six-fold blocked-CV scatter as reported | [M, §4.2, Table 5, App. A] | Spatial folds, log-ratio MAD formula, published scores. | Text and arithmetic definitions inspected; no arrays rerun. | **SUPPORTED** | No direct independent numerical replication. |
| Validated accuracy in highest-extinction GALEX-free plane | [M, §§4.2–4.4, 6] | Demonstrated mismatch and only indirect FIMS/UVOT constraints. | No independent target-region reference sampled. | **UNRESOLVED** | Need truly blind plane observations. |
| Nominal 68% SIGMA coverage in every hard stratum | [M, §4.6, Table 10] | FUV high-dust coverage 0.52–0.54. | Compared manuscript prose and table. | **CONTRADICTED** | Aggregate calibration remains near nominal; local coverage differs. |
| Provenance weights sum to one in output FITS | [M, eqs. 8–9; Table 7] | Convex/nested weight specification and classes. | Algebraic construction read, actual data not inspected. | **SUPPORTED** | Numerical array test unexecuted. |
| Every measured value is literally unchanged | [T, §3; M, §§3.2–3.4, 3.10] | Foreground removal, gain adjustments, local post-fusion corrections. | Cross-checked apparently absolute wording against exceptions. | **MISLEADING** | Measured imaging generally takes priority after processing. |
| External instruments independently verify all filled sky | [M, §4.4] | FIMS constrains large scales; UVOT partly supplies fusion. | Analyzed documented dependencies and low-weight comparisons. | **MISLEADING** | Some measurements provide partial independent checks. |
| Model-only fine structure establishes new UV discoveries | [M, §§3.10, 6; T, §3] | Predictor reuses external templates; authors expressly caution. | Checked source statements on valid science uses. | **CONTRADICTED** | New structure needs direct UV follow-up. |
| End-to-end public numerical archive independently accessible | [M, §7; T, §6] | Manifest and executable files described; download links promised. | Inspected current landing-page inventory and artifact-name searches. | **UNVERIFIED** | No stable public replay package obtained. |
| AI agents executed all described stages autonomously | [A, narrative] | Author reports parallel agents and human intervention. | Read first-party narrative; no machine logs examined. | **UNVERIFIED** | Author account is not an independent execution audit. |

**No row asserts VERIFIED numerical map performance:** source-text and arithmetic verification are not the same as independent observational validation.

## 6. Five priority technical risks and decisive tests

**Risk 1 — Domain shift in the actual missing Galactic plane (highest materiality).**
*Demonstrated limitation:* distribution separation AUC 0.83 FUV, importance-weight effective sample size 2.2%, high dust and stellar-density differences [M, §4.2, Appendix B]. *Scientific consequence:* favorable observed-sky CV may miscalibrate predicted brightness and errors in the target population. *Decisive test:* pre-registered, instrument-independent UV

observations sampled across E(B−V), stellar density, |b| and distance to GALEX, with bandpass and angular scale matched before scoring.

**Risk 2 — Calibration failure in conditional uncertainty and extreme tails.**
*Demonstrated limitation:* 0.52–0.54 rather than 0.68 FUV one-sigma coverage in E(B−V)=0.8–1.6 and extreme standardized tails [M, §4.6, Table 10]. *Consequence:* per-pixel intervals may be anti-conservative for some scientific decisions. *Decisive test:* independent 68%, 95.4%, and 99.73% spatially clustered coverage estimates by provenance, dust, brightness and distance, with SIGMA and coherent SIGMA_SYS handled distinctly.

**Risk 3 — Validation dependence and tuning optimism.**
*Documented design limitation, not established prohibited leakage:* cross-instrument FIMS constraint overlaps a validation comparator; the manuscript says hyperparameters were tuned with reporting folds and references were consulted during development, so no fully blind experiment exists [M, §§4.4, 6(v)]. *Consequence:* existing figures could be optimistic for fresh sky. *Decisive test:* freeze model and gains, withhold an entire instrument and separately obtain a completely new, hidden target-region sample; replay preprocessing at each fold boundary.

**Risk 4 — Apparent fine-scale detail without equivalent predictive resolution.**
*Demonstrated phenomenon:* predicted small-scale angular power is attenuated by 13–40%, while presentation uses an arcminute grid and the authors warn against discovery use [M, §§3.10, 4.7, 6; T, §5]. *Consequence:* visual users may mistake template-driven morphology for directly observed new structure. *Decisive test:* compare scale-dependent transfer functions and point-source/filament retrieval against withheld true images, with provenance overlay on all publication figures.

**Risk 5 — Missing public computation artifacts and incomplete raw-to-map replay.**
*Observed availability gap:* landing page does not presently link promised FITS/HiPS packages; manuscript notes interactive upstream stages [T, §6; M, §7]. *Consequence:* statistical claims and file-level integrity remain externally untested. *Decisive test:* publish immutable archive, file hashes, environments, pipeline source and intermediate products; then independently regenerate Table 5, Table 10, weights and one full-fusion mock-gap case.

A related **absolute-photometry** limitation deserves explicit attention for certain uses: inherited foreground zero-point systematics may reach approximately 100 CU and are not represented in the main per-pixel uncertainty columns [M, abstract; §4.6]. This can be more important than the five risks above when assessing absolute background levels or inferred extragalactic monopoles.

## 7. Suitability by scientific use

| Intended use | Assessment | Conditions and exclusions |
|---|---|---|
| Education and contextual all-sky visualization | **Suitable with disclosure** | Show a visible provenance/resolution legend; avoid calling reconstructed pixels observations. |
| Large-scale FUV surface-brightness planning | **Conditionally useful** | Use all classes with SIGMA/SIGMA_SYS and FIMS calibration bracket; account for distance and dust. |
| UV mission field selection and exposure planning | **Conditionally useful** | Treat bright-source and modeled foregrounds as estimates, not hard constraints; inspect crowded-plane uncertainties. |
| High-resolution diffuse photometry and structural power | **Restricted to strongly imaging-dominated pixels** | Manuscript recommends `W_GALEX>0.99`; preserve transfer-function and zero-point caveats. |
| FUV/NUV colour, FUV-specific gas-line analysis | **Direct-imaging dominated bands required** | Do not use NUV-transferred FUV as an independent observed FUV signal. |

| Intended use | Assessment | Conditions and exclusions |
|---|---|---|
| Absolute extragalactic background/monopole | **Not established by this reconstruction alone** | UV-BKGD zero point and correlated additive systematics are inherited. |
| New small-scale source or filament discovery in model-only regions | **Not justified** | Predictor reuses dust/starlight templates and may not contain independent novel UV information. |

This use taxonomy follows the product's documented provenance recommendations rather than declaring the reconstructed area scientifically worthless [M, §3.10, Appendix D].

# 8. Reproducibility status and minimum independent benchmark

## 8.1 Availability at review cut-off

| Required material | What the public record demonstrates | This review |
|---|---|---|
| Anthropic narrative, technical site and 41-page manuscript | Publicly readable [A, T, M] | **Read and cross-checked** |
| Public base mission catalogues and source surveys | Public observational archives described and linked by the author [M, Table 2; A] | **Custodians identified; raw archives not ingested** |
| Ménard's fused FUV/NUV FITS maps, weight/provenance arrays | Specified by [M, Table 7], links promised by [T, §6] | **Not downloaded; availability not established** |
| Full `manifest.json` with SHA-256 checksums | Specified by [M, §7] | **Not obtained; no published file hash verified** |
| `gapfill_benchmark_nside512.npz`, out-of-fold arrays, `score_benchmark.py` | Specified by [M, §7, Appendix A] | **Not obtained or executed** |
| `fusion.py`, `run_fusion_{fuv,nuv}.py`, `pipeline_code.tar.gz`, environment | Described by [M, §7, Table 7] | **Not obtained or executed** |
| New high-extinction GALEX-free UV test observations | Not provided as a blind representatively sampled benchmark | **Not performed** |

## 8.2 Smallest executable independent replay

A meaningful **computational** check should first retrieve and authenticate the benchmark and source manifest. Freeze exact file versions and SHAs, reproduce `score_benchmark.py` on the six-fold `nside=512` out-of-fold predictions, implement an independent weighted-median/MAD scorer, and compare the Table 5 total-intensity matched values of 0.060±0.002 dex (FUV) and 0.060±0.001 dex (NUV) [M, §4.2, Table 5, Appendix A]. Any acceptance tolerance must be specified *before* scoring: for example **≤0.0005 dex** numerical agreement on identical archived arrays and weights is a **proposed audit criterion**, not a demonstrated match; published bootstrap uncertainties are substantially larger and concern statistical sampling, not numerical replay tolerances.

Next recompute the diffuse-target and total-target results **separately**, with labelled equalized versus pre-equalized reference arrays; compare the 0.052/0.058 and 0.057/0.060 total-target definitions rather than averaging them. Stratify by E(B−V), |b|, Gaia source density, Hα, provenance class and GALEX separation. Verify weight sum-to-one, dominant-code `argmax`, the strict FUV measured subset, and SIGMA/SIGMA_SYS coverage. Rebuild at least one end-to-end mock gap, withholding its observations **before** any dependent calibration, parameter selection or source-feature operation.

Those steps could establish numerical repeatability of reported benchmark and fusion outputs. They **cannot alone** establish accuracy in the target domain that has no GALEX observations. That requires an independent blind observational

sample; the audit should record the new instrument's passband transformation and error model before declaring a comparison decisive.

## 9. Direct answers and final disposition

**What did the project establish?** A documented methodology for combining measured, calibrated, externally constrained and predicted UV intensities into a seamless two-band all-sky product, together with numerous internal quantitative tests and meaningful provenance and uncertainty specifications. This review independently checked descriptions and the existence of earlier Jo et al. FUV work, not the numerical output files.

**What remains supported but not independently verified?** The reported sky fractions, star count, robust errors, seam metrics, uncertainty coverage and implementation behavior. No end-to-end replay of the claimed production files occurred.

**What is scientifically unresolved?** Actual prediction and uncertainty performance in the highest-extinction, source-crowded, most distant GALEX-free plane; precise explanation of the 110-versus-zero high-dust GALEX statements; certain similarly named total-target/coverage statistics; and full artifact integrity.

**Where do public descriptions require qualification?** The unqualified historical “first complete UV map”; the compression of non-uniform log-scatter into “about 10%”; the implication that all measured pixels are literally unchanged; and any interpretation of the seamless render as homogeneous-resolution observational truth. The project website and manuscript already provide many of the needed cautions, so the criticism is targeted rather than global.

**What would most improve the evidence?** First, release verifiable numerical artifacts, benchmarks, code and provenance files. Then test a frozen map against independently collected, geographically and physically representative UV observations of high-extinction gaps, reporting error and coverage by difficulty stratum.

**Final scientific judgment:** The map is a credible and unusually well-documented reconstruction of a heterogeneous observational record. Its reported observed-sky benchmark performance is credible but unreplicated here, and does not constitute independent proof of comparable accuracy in the unobserved Galactic plane. The appropriate next step is reproducible release and target-domain validation, not dismissal of the project or unconditional acceptance of the visual result.

## Primary-source references and audit scope

**[A]** Ménard, B., “The missing map of the sky,” *Anthropic*, October 8, 2026. https://www.anthropic.com/research/the-missing-map-of-the-sky
**[T]** Ménard, B., “The full sky in the ultraviolet,” project website, October 2026; sections 2–7 reviewed October 9, 2026. https://menard.pha.jhu.edu/uvmap/
**[M]** Ménard, B., *Full-sky ultraviolet maps from harmonised GALEX, Swift-UVOT, FIMS/SPEAR, TD-1 and Gaia data*, **Draft, October 2026**, 41 PDF pages; citation “p.” refers to the PDF page number (not a journal page). https://menard.pha.jhu.edu/uvmap/docs/uvmap_manuscript.pdf
**[J]** Jo, Y.-S., Choi, Y.-J., Kim, M.-G., Woo, C.-H., Min, K.-W. & Seon, K.-I. (2021), “Construction of a far-ultraviolet all-sky map from an incomplete survey: application of a deep learning algorithm,” *Monthly Notices of the Royal Astronomical Society*, **502**, 3200–3209, doi:10.1093/mnras/stab066. https://academic.oup.com/mnras/article/502/3/3200/6095714

**Audit boundary:** Primary literature and public descriptions were read and internally compared. Prior draft audits supplied in the conversation informed the questions but were not treated as measurements or independent external corroboration. No released scientific FITS, NPZ or executable pipeline was obtained or independently run; no new observations were collected; no human astrophysical peer review is claimed. Report version 1.0 is subject to revision if the manuscript or project releases change.

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